Name: ______________________

Class: ______________________

Date: ______________________

Mr. Rawson • IB DP Physics

C.1 – Key Elements

Answer Key

Almost nothing on this page is a question on its own. All of it is needed to answer questions about something else, which is exactly why it is worth knowing cold. Every idea here is tied to the spring you timed on Monday.

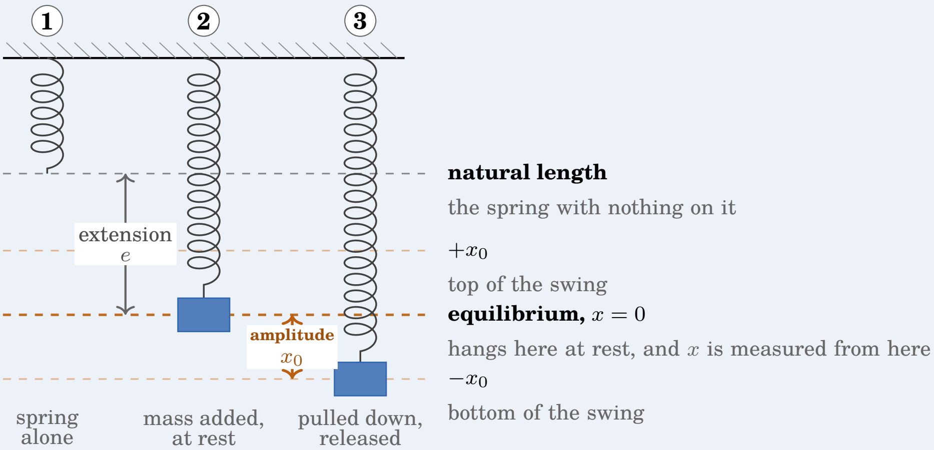
IB Physics Guide • §C.1 Simple harmonic motion • ***Theme C, Wave Behaviour***

C1-1 the conditions that lead to simple harmonic motion
C1-2 the defining equation $a = -\omega^2 x$
C1-3 describing an oscillator by period, frequency, angular frequency, amplitude, equilibrium position and displacement
C1-4 $T = \frac{1}{f} = \frac{2\pi}{\omega}$
C1-5 $T = 2\pi\sqrt{m/k}$
C1-6 $T = 2\pi\sqrt{l/g}$

Where you measure from

Key concept

Read the three springs left to right. Each one adds a single idea to the one before it, and the word for each is beside it.



The vocabulary, in the guide's own terms (C.1: *a particle undergoing SHM can be described using time period, frequency, angular frequency, amplitude, equilibrium position and displacement*)

equilibrium position	—	the position at which the resultant force on the particle is zero, so it stays there at rest
displacement	x	the distance in a stated direction from the equilibrium position, so it carries a sign
amplitude	x_0	the maximum magnitude of the displacement from the equilibrium position

The two lengths a hanging spring has at once

Your spring had a natural length before you hung anything on it. Adding the mass stretched it to a new resting position, and the oscillation happens about *that*, not about the natural length.

So a hanging mass has two different measurements at every instant, the **extension from the natural length**, which is what stores elastic energy, and the **displacement from equilibrium**, which is what appears in $a = -\omega^2 x$. They are never equal, and a question that mixes them up is usually testing whether you will.

The trap that costs a whole calculation

On the figure above, $+x_0$ and $-x_0$ are the two extremes and $x = 0$ sits between them, so the amplitude is **half** the full swing. If a question gives you the distance from the lowest point to the highest point, halve it before you use it anywhere.

Your own evidence. You pulled the mass down by three different amounts and timed each. The period did not change. Amplitude is the one quantity the period of this oscillator does not care about.

The restoring condition, and the minus sign

Key concept

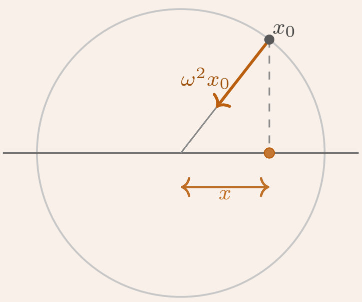
$$a = -\omega^2 x$$

Read it as a sentence. The acceleration is **proportional to the displacement from equilibrium**, and it is **directed towards equilibrium**, which is what the minus sign says. Both halves are needed, and in a two-mark question they are the two marks.

$a \propto -x$ and $F \propto -x$ say the same thing, so either earns the mark provided your words explain what the sign means. Saying the acceleration is negative earns nothing on its own.

Where $a = -\omega^2 x$ comes from

Watch something going round a circle from the side, with no depth perception, and what you see is an oscillation. That is the whole idea, and the algebra is four short steps.



the shadow on the diameter is the oscillation

$$a_c = \frac{v^2}{r} \quad \text{going round a circle}$$

$$v = \omega r \quad \text{speed round that circle}$$

$$a_c = \frac{(\omega r)^2}{r} = \omega^2 r \quad \text{substitute}$$

$$\vec{a} = -\omega^2\, \vec{r} \quad \text{it points back at the centre}$$

That last line is a statement about vectors, so it is true of the components too. The shadow on the diameter has displacement x, and its acceleration is the shadow of $\vec{a}$, which gives

$$\boxed{a = -\omega^2 x}$$

The minus sign is not decoration. It says the acceleration is a vector pointing the opposite way to the displacement, which is what makes the motion come back rather than run away.

This one borrows $a_c = v^2/r$ from Theme A. If you have not met it yet, take it on trust for now and come back *when you have met circular motion in Theme A, the rest of the sheet does not depend on it.*

Three ways of saying how fast it repeats

Key concept

$$T = \frac{1}{f} = \frac{2\pi}{\omega} \qquad \text{so} \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

T in seconds, f in hertz, ω in radians per second. Moving between these three is asked about directly only occasionally, and needed in more than half of all C.1 questions, usually as the first line of a longer answer.

The two standard oscillators

Key concept

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad\qquad T = 2\pi\sqrt{\frac{l}{g}}$$

Neither period depends on amplitude. The pendulum's period does not depend on its mass either, which surprises most people the first time.

Turning your graph into a spring constant

Squaring the spring result gives $T^2 = \dfrac{4\pi^2}{k}\,m$, a straight line through T^2 against m with

$$\text{gradient} = \frac{4\pi^2}{k} \qquad \text{so} \qquad k = \frac{4\pi^2}{\text{gradient}}$$

Straightening a relationship before you draw it is worth doing whenever you can, because a straight line is the only shape the eye judges honestly.

Your line does not pass through the origin, and that is real rather than sloppy. The spring has mass of its own which oscillates too, so a small positive intercept appears. It lifts the line without changing the gradient, so your value of k survives it.

Higher level only

Phase angle ϕ says where in the cycle the oscillator was at $t = 0$.

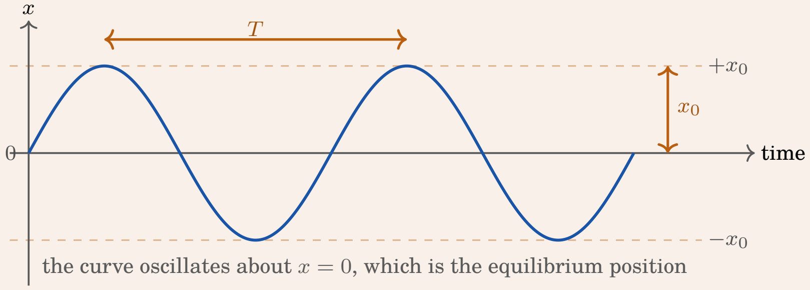
$$x = x_0 \sin(\omega t + \phi) \qquad v = \omega x_0 \cos(\omega t + \phi) \qquad v = \pm\,\omega\sqrt{x_0^2 - x^2}$$

$$E_T = \tfrac{1}{2} m\omega^2 x_0^2 \qquad E_p = \tfrac{1}{2} m\omega^2 x^2$$

ωt and ϕ are in **radians**, always. Put the calculator in radians before you start and leave it there, because $\sin(2.4)$ in degree mode returns an answer roughly sixteen times too small and looks entirely reasonable.

The same three words, on a graph

What you will plot tomorrow



Amplitude is measured from the equilibrium line up to a peak, never from the bottom of the curve to the top. Period is the time from any point to the next point doing the same thing, so peak to peak, or trough to trough, or one rising zero to the next rising zero. Reading it between a peak and a trough gives you half a period, and it is the commonest way to lose the mark.

Your own oscillator, from Monday

Fill this in once and keep it. You will need these numbers for the write-up, and they are worth more to you than any example I could invent.

Gradient of T^2 against m ____________________ $\mathrm{s^2\,kg^{-1}}$

Spring constant $k = 4\pi^2/\text{gradient}$ ____________________ $\mathrm{N\,m^{-1}}$

Intercept, and whether it is positive ____________________

Uncertainty in your timing, ΔT ____________________ s

Key concept

Cover this sheet and see whether you can still say: where displacement is measured from, what amplitude is measured between, what the minus sign in $a = -\omega^2 x$ means, and how to get from T to ω. Anything you cannot say without looking is the thing to read again, and it will be the thing costing you marks in a fortnight.

Quick checks

Short, and none of them needs more than a line or two. Do these before you start the problem set, they are the ideas the rest of it stands on.

1) State A spring hangs with a natural length of 20.0 cm. A mass is attached and the spring hangs at rest at 26.0 cm. The mass is pulled down to 31.0 cm and released.

[4] Give the length at the equilibrium position, the amplitude, the displacement at the moment of release, and the extension from the natural length at that same moment.

Mark scheme

Equilibrium is at **26.0 cm**, where it hung at rest. Amplitude is $31.0 - 26.0 =$ **5.0 cm**. Displacement at release is **5.0 cm** below equilibrium, so -5.0 cm if downwards is taken as negative. Extension from the natural length is $31.0 - 20.0 =$ **11.0 cm**. *The point of the question is that 5.0 and 11.0 are both correct answers to different questions.*

2) Calculate An oscillator moves 9.0 cm from its lowest point to its highest point. State its amplitude.

[1]

Mark scheme

$x_0 = 9.0/2 =$ **4.5 cm**. The figure given is the full swing, and amplitude is measured from equilibrium to one extreme.

3) Calculate A mass on a spring has a period of 0.84 s. Find its frequency and its angular frequency.

[2]

Mark scheme

$f = 1/T = 1/0.84 =$ **1.19 Hz**. $\omega = 2\pi/T = 2\pi/0.84 =$ **7.48 rad s^{-1}**.

4) Determine A graph of T^2 against m has a gradient of 2.6 s^2 kg^{-1}. Find the spring constant.

[2]

Mark scheme

$k = 4\pi^2/\text{gradient} = 39.48/2.6 =$ **15 N m^{-1}** (2 s.f.). Check the units, s^{-2} kg is the same as N m^{-1}.

5) Explain An object accelerates according to $a = -40x$, with x in metres. Explain whether this is simple harmonic motion, and find the period.

[3]

Mark scheme

It is SHM. The acceleration is proportional to the displacement from equilibrium and the minus sign puts it towards equilibrium, which is the defining condition. Comparing with $a = -\omega^2 x$ gives $\omega^2 = 40$, so $\omega = 6.32$ rad s^{-1} and $T = 2\pi/\omega =$ **0.99 s**.

6) Explain Your graph of T^2 against m did not pass through the origin. Explain why that is expected.

[2]

Mark scheme

The spring has mass of its own, and part of it oscillates with the load, so the system behaves as though the hanging mass were slightly larger. That adds a small positive intercept. It shifts the line up rather than tilting it, so the gradient, and therefore k, is unaffected.

Higher level only

7) State An oscillator is described by $x = 0.050\sin(12t + \pi/3)$, in metres and seconds. Give the amplitude, the angular frequency, the period, and the displacement at $t = 0$.

[4]

Mark scheme

$x_0 =$ **0.050 m**, $\omega =$ **12 rad s^{-1}**, $T = 2\pi/12 =$ **0.52 s**. At $t = 0$, $x = 0.050\sin(\pi/3) = 0.050 \times 0.866 =$ **0.043 m**. In degree mode this returns 0.0009 m, which is the mistake to watch for.